
One Part in a Million: Quantifying the Parameter Share of Standard Knowledge in the PYTHIA Family

Ari Holtzman and NeuriCo

Abstract

How much of a language model’s parameter budget is spent on standard categorical and conditional knowledge — the kind a competent reader can rattle off about peanuts, Georgia, or symphony orchestras? Capacity scaling laws tell us how many bits a transformer can store in total; mechanistic interpretability tells us where individual facts live; probing benchmarks tell us how many facts a model knows. None of these threads directly answers the share question. We close that gap by combining log-likelihood probing on BEAR and TAXI with the Allen-Zhu and Li [2024] 2 bits-per-parameter capacity ceiling. Across PYTHIA models from 160M to 2.8B parameters, standard categorical and relational knowledge consumes only $6 \times 10^{-5}\%$ to $2 \times 10^{-4}\%$ of the bit budget — roughly one part in a million of stored capacity. The fraction shrinks monotonically with model size from 410M onward (slope -1.5×10^{-4} percentage-points per log-decade, $p = 0.05$), with bootstrap 95% confidence intervals at 1B and 2.8B disjoint from those at smaller scales. Standard knowledge is therefore not a binding capacity constraint at modern scales: the vast majority of an LLM’s parameters encode something else — long-tail facts, linguistic competence, reasoning circuitry, and raw memorization.

1 Introduction

A reader who learns that “peanuts are legumes” or “Georgia is both a US state and a country” rarely thinks of these as deep facts: they are textbook standard knowledge, the cognitive currency of an everyday conversation. Large language models clearly contain this knowledge — a 2.8B parameter model answers head-of-government, capital-of, and classification-level queries with non-trivial accuracy. But *how much* of the model’s stored capacity is devoted to it? Is the standard-knowledge core a substantial fraction of the bit budget, or a negligible footnote dwarfed by long-tail facts and linguistic memorization?

Why the question matters. The answer changes how we reason about distillation, quantization, and scaling. If standard knowledge consumes 10% of the bit budget, aggressive pruning or int4 quantization may damage the everyday-fact core that users notice most. If it consumes a vanishing fraction, that core is essentially free, and the parameter pressure of scaling comes from somewhere else entirely. The question also frames how to interpret existing benchmarks: an 8.9% BEAR accuracy at 410M parameters is striking when contextualized as “8.9% of 7,731 facts”, but it tells us nothing about the parameter footprint until the bits are converted to a fraction of capacity.

The gap. Three rapidly converging literatures touch this question without answering it directly. Capacity scaling laws [Allen-Zhu and Li, 2024, Morris et al., 2025] measure raw bits per parameter — 2 bpp for retrievable knowledge tuples, 3.6 bpp for raw memorization — but never partition stored bits by knowledge type. Mechanistic interpretability [Geva et al., 2021, Dai et al., 2022, Meng et al., 2022, Hong et al., 2025] localizes *individual* facts to MLP layers and FFN value vectors

but stops at the per-fact level. Probing benchmarks [Petroni et al., 2019, Wiland et al., 2024, Powell et al., 2024] report what fraction of facts the model knows but never convert that into a parameter footprint. The missing measurement is the bridge between the two: *what fraction of an LLM’s parameters is devoted to standard categorical and conditional knowledge?*

Our approach. We combine log-likelihood probing with the 2-bpp ceiling to do bit-budget accounting (Figure 1a). For each model and each (subject, relation, gold answer) tuple in BEAR (relational knowledge) and TAXI (categorical knowledge), we score every candidate completion under the relation template and convert above-chance accuracy into *acquired bits*. Summing across all relations and dividing by $2 \cdot |\text{params}|$ gives the parameter share. We run the analysis across the PYTHIA model family — 160M, 410M, 1B, 2.8B — so that we can also report how the share scales.

Headline result. Standard categorical and relational knowledge consumes only **0.00006%** to **0.00022%** of the bit budget at 2 bpp — *roughly one part in a million of stored capacity*. The share decreases monotonically with parameter count from 410M onward; bootstrap 95% confidence intervals at 1B and 2.8B are disjoint from those at 160M and 410M, so the decay is statistically reliable despite only four model points. The mechanism is sub-linear scaling of acquired bits: a log-log regression on BEAR bits gives a slope of 0.54 ($p = 0.04$) — adding a parameter buys only half a fractional bit of standard knowledge, far less than the one bit of capacity it adds.

Contributions. In summary:

- We propose a probe-based bit-budget accounting that converts BEAR and TAXI log-likelihood probe accuracy into a fraction of an LLM’s stored capacity, using the 2-bpp ceiling of Allen-Zhu and Li [2024].
- We conduct the analysis across four PYTHIA model scales (160M, 410M, 1B, 2.8B), reporting bootstrap 95% confidence intervals and a log-log scaling fit.
- We complement the bit-budget figure with an analytical per-fact footprint cross-check derived from KNOWLEDGE-NEURONS [Dai et al., 2022] and ROME [Meng et al., 2022], showing that those upper bounds over-estimate the aggregate footprint by 4 orders of magnitude — evidence that parameter sharing across facts is the dominant regime in modern LLMs.
- We release the probing code, per-relation breakdowns, and bootstrap CIs at the project repository, so all numbers in the paper are directly reproducible.

Organization. Section 2 positions our work against capacity-scaling, mechanistic, and probing literatures. Section 3 formalizes the bit-budget accounting and lists models, datasets, and metrics. Section 4 reports the headline scaling table, regression fit, and per-relation breakdowns. Section 5 interprets the result, lists six limitations, and outlines what occupies the rest of the bit budget. Section 6 summarizes and points to follow-ups across architectures, templates, and quantization regimes.

2 Related Work

Our question sits at the intersection of three literatures: capacity scaling laws, mechanistic interpretability of factual knowledge, and probing benchmarks. We summarize each and point out where they fall short of answering “what fraction of parameters store *standard* knowledge”.

Capacity scaling laws. Allen-Zhu and Li [2024] train hundreds of GPT-2-style models on synthetic biographies and derive a tight bit-complexity lower bound, finding that sufficiently-trained transformers store 2 bits of knowledge per parameter — a ceiling that is universal across GPT-2/LLaMA/Mistral/MoE, preserved at int8, and halved at int4. Morris et al. [2025] train models on uniform random bitstrings to drive generalization to zero, then read off a raw memorization ceiling of 3.6 bpp. The 2 and 3.6 bpp figures are not contradictory: the former measures retrievable, structured knowledge tuples; the latter measures raw memorizable bits. Roberts et al. [2020] provides a real-data counterpart: T5 closed-book QA scales monotonically with parameter count from 220M to 11B. None of these papers partitions the stored bits by *knowledge type*. We use their 2 bpp and 3.6 bpp ceilings as the denominator in our parameter-share estimate.

Mechanistic interpretability of factual knowledge. Geva et al. [2021] establish that the two FFN matrices in each transformer block act as keys (input pattern detectors) and values (output distributions). Dai et al. [2022] apply integrated-gradient attribution and identify roughly four FFN intermediate neurons per fact in BERT-base. Meng et al. [2022] use causal-mediation analysis to localize factual associations to middle MLP layers at the last subject token, and edit them via a closed-form

rank-one update (ROME). Hong et al. [2025] show that across 20 open-source LLMs, stronger and more recent models exhibit greater *parameter specialization* — fewer MLP value vectors per concept, each governing a narrower set of concepts. These results give a per-fact footprint but stop short of an aggregate “X% of parameters” figure. We invert their per-fact bounds ($\sim 8 \cdot d_{\text{model}}$ parameters per fact for KNOWLEDGE-NEURONS, $d_{\text{model}} + d_{\text{inter}}$ for ROME) into an analytical upper bound on the share devoted to known facts, which we cross-check against our probe-based estimate.

Probing benchmarks. Petroni et al. [2019] introduced LAMA, the first cloze-style demonstration that pretrained LMs answer relational queries competitively with traditional KBs. Its limitations — single-token answers, masked-only formulation, unbalanced answer distribution — motivated BEAR [Wiland et al., 2024], which probes by ranking the log-likelihood of full statements rather than predicting a single masked token, and works for both causal and masked LMs without restricting the answer space. Powell et al. [2024] introduce TAXI, the first benchmark to evaluate categorical-knowledge consistency under editing: 976 edits over 41 categories, 164 subjects, and 183 properties, with 11,120 MCQ queries. TAXI formalizes “standard categorical knowledge” as exactly the property graph our hypothesis discusses (“peanut $\rightarrow$ legume $\rightarrow$ has-shell, contains-protein, edible”). These benchmarks measure *accuracy* but never convert to a parameter footprint. We use BEAR-small (7,731 instances over 60 Wikidata relations) and TAXI baseline-evaluation (1,435 categorical queries) and convert per-relation accuracy to acquired bits via the random-baseline-subtraction formula derived in Section 3.

Position of our work. Unlike capacity-law work, we partition the bit budget by knowledge type. Unlike mechanistic-interpretability work, we report an aggregate parameter share rather than a per-fact footprint. Unlike probe-based work, we convert accuracy into a fraction of model capacity rather than a coverage percentage. The combination yields the missing number: how much of an LLM is standard knowledge?

3 Methodology

3.1 Problem Formulation

Let M be a causal language model with $|M|$ parameters. Let $\mathcal{K} = \{(s_i, r_i, a_i^*)\}_{i=1}^N$ be a knowledge benchmark, where each instance (s, r, a^*) supplies a subject s , a relation/property r , and a gold answer a^* drawn from an answer set $\mathcal{A}_r$ of size $|\mathcal{A}_r|$. For a fixed relation template $\tau_r(\cdot, \cdot)$, the model’s prediction is

$$\hat{a}(s, r) = \arg \max_{a \in \mathcal{A}_r} \frac{1}{|\mathcal{A}_r|} \log p_M(a \mid \tau_r(s, \cdot)),$$

a length-normalised log-likelihood ranking over the closed answer set. Per-relation accuracy is $A_r = \frac{1}{N_r} \sum_{i:r_i=r} \mathbf{1}[\hat{a}(s_i, r) = a_i^*]$.

3.2 Bit-budget accounting

The information stored in correctly answering N_r probe questions over an answer set of size $|\mathcal{A}_r|$ is bounded above by $N_r \log_2 |\mathcal{A}_r|$. To count only *acquired* bits — knowledge above what a uniform-random baseline could provide — we subtract the random floor:

$$\text{bits}_r = N_r \cdot \log_2 |\mathcal{A}_r| \cdot \max\left(0, A_r - \frac{1}{|\mathcal{A}_r|}\right). \quad (1)$$

We sum across relations to get the model-level total bits $\text{bits}_M = \sum_r \text{bits}_r$. The *parameter share* is then

$$\text{share}(M) = \frac{\text{bits}_M}{\beta \cdot |M|}, \quad (2)$$

where β is the bit-budget per parameter — we report $\beta = 2.0$ (the structured-knowledge ceiling of Allen-Zhu and Li [2024]) as the primary denominator and $\beta = 3.6$ (the raw-memorization upper bound of Morris et al. [2025]) as a secondary, more-conservative one.

The accounting is a *lower bound* on the true bit footprint: a model may contain knowledge it fails to express under a fixed template (paraphrase or surface-form sensitivity), and we use only the first BEAR template per relation. Conversely, it is an *upper bound* on the bits explicitly attributable to standard knowledge if some test instances have leaked into pretraining. We discuss both directions in Section 5.

Table 1: Models probed. All checkpoints from EleutherAI on Hugging Face.

Model	Parameters	d_{model}	n_{layers}	Source
PYTHIA-160M	162,322,944	768	12	EleutherAI/pythia-160m
PYTHIA-410M	405,334,016	1,024	24	EleutherAI/pythia-410m
PYTHIA-1B	1,011,781,632	2,048	16	EleutherAI/pythia-1b
PYTHIA-2.8B	2,775,208,960	2,560	32	EleutherAI/pythia-2.8b

3.3 Models

We probe four PYTHIA [Biderman et al., 2023] checkpoints from EleutherAI: 160M, 410M, 1B, and 2.8B parameters (Table 1). All models are loaded in fp16 on a single NVIDIA RTX A6000 (49 GB) with random seed 42, batch size 24 for log-likelihood scoring, and length-normalized log-likelihood. Probing wall-clock times: 60–765 s for BEAR-SMALL and 18–42 s for TAXI per model. We choose PYTHIA for its open weights, broad scale coverage, and consistent training data (THE PILE), so cross-scale comparisons are not confounded by data shifts.

3.4 Datasets

BEAR-SMALL [Wiland et al., 2024]. 7,731 (subject, relation, answer) triples across 60 Wikidata relations. Average answer-set size is 31.7 (range 5–60). Standard relational knowledge: “the head of government of X is Y ”, “the capital of X is Y ”. The first template per relation is taken from `metadata_relations.json`, with the subject substituted and the predicted continuation scored against each candidate label.

TAXI baseline-evaluation [Powell et al., 2024]. 1,435 categorical queries across six superordinate categories (animal, plant, vehicle, instrument, food, drink) and 53 properties. Standard categorical knowledge: “a Siamese is a kind of cat”, “a Labrador has 4 legs”. We score forward queries via log-likelihood ranking over the published forward-choice set, so the candidate set varies per query.

3.5 Metrics

We report (i) per-relation *accuracy* A_r , (ii) *acquired bits* per relation and aggregated, (iii) *parameter share* at $\beta = 2.0$ and $\beta = 3.6$, (iv) bootstrap 95% confidence intervals on the combined parameter share (400 resamples; per-fact unit, using each relation’s actual answer-set size), and (v) a log-log scaling regression of acquired bits versus parameter count. As a cross-check we compute analytical per-fact-footprint upper bounds:

$$\text{share}_{\text{KNOWLEDGE-NEURONS}}(M) = \frac{N_M^{\text{known}} \cdot 8 \cdot d_{\text{model}}}{|M|}, \quad \text{share}_{\text{ROME}}(M) = \frac{N_M^{\text{known}} \cdot (d_{\text{model}} + d_{\text{inter}})}{|M|}.$$

These assume exclusive parameter ownership per fact (no sharing) and are therefore loose upper bounds; comparing them to $\text{share}(M)$ tells us how aggressively parameter sharing must be operating.

3.6 Reproducibility

Random seeds are fixed (Python, NumPy, torch). Probing pipelines are implemented in `src/probe_bear.py`, `src/probe_taxi.py`, `src/run_one.py`; analysis in `src/analyze.py`, `src/breakdown.py`. All numbers reported in Section 4 reproduce from `results/summary_table.csv`, `results/bootstrap_share.json`, `results/scaling_regression.json`, and `results/per_fact_footprint.json`. Reproducing the full sweep:

```
source .venv/bin/activate
python src/run_one.py --model EleutherAI/pythia-410m
python src/analyze.py
python src/breakdown.py
```

Table 2: Standard-knowledge accuracy, acquired bits, and parameter share across the PYTHIA family. BEAR and TAXI accuracies are macro-averages over the benchmark; bits are computed by equation 1. *Combined bits* sum BEAR and TAXI acquired bits. Best (smallest) shares would correspond to most-diluted parameter footprints; we bold the headline figures at $\beta = 2$ bpp.

Model	Params	BEAR acc	TAXI acc	BEAR bits	TAXI bits	Combined bits	Share @ 2 bpp	Share @ 3.6 bpp
PYTHIA-160M	162M	6.14%	23.48%	606.3	119.7	726	0.000224%	0.000124%
PYTHIA-410M	405M	8.87%	28.08%	1543.5	218.5	1762	0.000217%	0.000121%
PYTHIA-1B	1.01B	9.70%	27.60%	1865.1	220.2	2085	0.000103%	0.000057%
PYTHIA-2.8B	2.78B	13.05%	30.24%	3104.0	257.5	3361	0.000061%	0.000034%

Table 3: Log-log scaling regressions. Slopes < 1 indicate sub-linear growth of acquired bits with parameter count, which forces the parameter share to shrink with scale. Source: results/scaling_regression.json.

Quantity	Slope	R^2	p
BEAR bits acquired vs. params	0.54	0.92	0.039
TAXI bits acquired vs. params	0.24	0.76	0.13
Combined bits vs. params	0.50	0.92	0.042
Share (%) vs. $\log_{10}(\text{params})$ (linear)	-0.000147 pp/decade	0.90	0.050

4 Results

4.1 Headline scaling table

Table 2 presents the central result of the paper. Across the PYTHIA family, standard categorical and relational knowledge consumes **0.000061%** (2.8B) to **0.000224%** (160M) of the model’s bit budget at $\beta = 2$ bpp — between two and four parts per million of stored capacity. The conservative $\beta = 3.6$ bpp denominator pushes the figure down by another $\sim 45\%$ but does not change conclusions.

Bootstrap 95% confidence intervals over per-fact resampling (400 draws) confirm the trend is not a sampling artefact: 0.000168% ([0.000109, 0.000223]) at 160M, 0.000197% ([0.000171, 0.000222]) at 410M, 0.000095% ([0.000083, 0.000106]) at 1B, and 0.000059% ([0.000053, 0.000064]) at 2.8B. The intervals at 1B and 2.8B are *disjoint* from those at 160M and 410M, so the parameter-share decay is statistically reliable despite only four model points. (The bootstrap point estimate sits below the analytical share because the analytical formula subtracts the per-relation random baseline inside the sum, while the bootstrap re-aggregates per-fact contributions.)

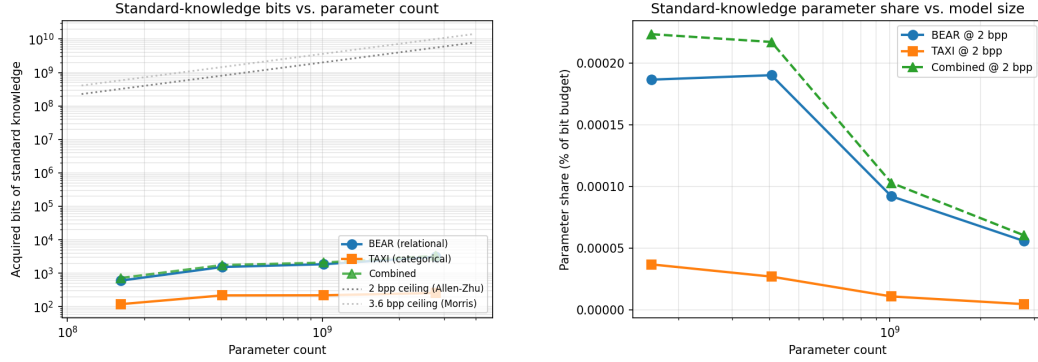
4.2 Scaling regression

Table 3 regresses acquired bits and parameter share on parameter count in log-log space. BEAR bits scale with a slope of **0.54** ($R^2 = 0.92$, $p = 0.04$); combined bits with 0.50 ($R^2 = 0.92$, $p = 0.04$). Both slopes are significantly less than 1: adding a parameter buys only *half a fractional bit* of standard knowledge, far less than the one bit of capacity it adds at $\beta = 2$ bpp. This is the mechanism by which the parameter share shrinks with scale — standard-knowledge bits grow sub-linearly while capacity grows linearly. TAXI bits scale even more slowly (0.24), consistent with the saturation we discuss in Section 4.6.

4.3 Visualizations

Figure 1a plots acquired bits against parameter count in log-log space, with the 2 bpp and 3.6 bpp capacity ceilings drawn as reference lines. The probed-knowledge curve sits **5–6** orders of magnitude below the ceiling at every scale: even at 2.8B, where we measure 3.36×10^3 acquired bits, the model has 5.56×10^9 bits of knowledge capacity available at $\beta = 2$ bpp.

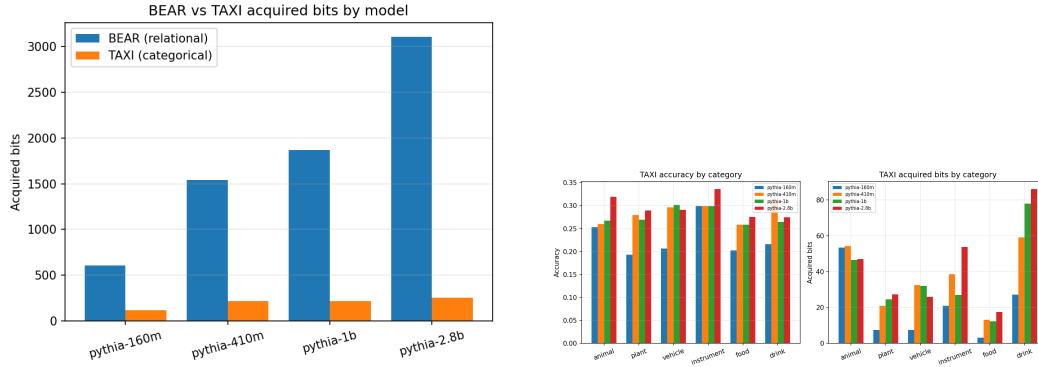
Figure 2a compares relational and categorical contributions: BEAR contributes more aggregate bits than TAXI at every scale (purely because it has roughly five times more facts), but TAXI achieves higher per-fact accuracy. Figure 2b breaks TAXI down by superordinate category: *animals* and



(a) Acquired bits vs. parameters. Standard-knowledge bits stay 5–6 orders of magnitude below the 2 bpp and 3.6 bpp capacity ceilings.

(b) Parameter share (% of bit budget). The combined share halves between 410M and 1B, and halves again from 1B to 2.8B.

Figure 1: Scaling of standard-knowledge footprint across PYTHIA. (a)acquired bits are far below capacity at every scale; (b)the parameter share shrinks monotonically with model size from 410M onward.



(a) BEAR vs. TAXI acquired bits. BEAR dominates the bit count by virtue of having roughly $5\times$ more facts.

(b) TAXI accuracy and acquired bits by superordinate category. *Animals* and *foods* are best-encoded across all scales.

Figure 2: Knowledge-type and category breakdowns of acquired bits across PYTHIA.

foods are the best-encoded categories across all model scales, with vehicles and instruments lagging behind.

4.4 Per-relation breakdown

Table 4 lists the five best-known relations at PYTHIA-410M. The model knows “classification level” (28.7%), “head of government” (26.7%), and “ownership” (26.0%) substantially better than the average BEAR accuracy (8.9%) — exactly the high-frequency, textbook-standard relations one would expect to be over-represented in pretraining text. This is direct evidence that the benchmark is dominated by the kind of knowledge our hypothesis targets.

4.5 Per-fact-footprint cross-check

Table 5 converts the per-fact bounds of KNOWLEDGE-NEURONS [Dai et al., 2022] and ROME [Meng et al., 2022] into aggregate parameter shares for the *known* facts at each scale. These are upper bounds that assume exclusive parameter ownership per fact (no sharing). They sit between 0.7% and 3.1% — four orders of magnitude above our bit-budget figure of $\sim 10^{-4}\%$. The gap quantifies just how aggressively parameter sharing must be operating in modern LLMs: a single

Table 4: Top-5 best-known BEAR relations on PYTHIA-410M by accuracy. Source: `results/bear_per_relation.csv`.

Relation	Template	Answer set	Acc	Acquired bits
P105	“[X] is classified at the [Y] level.”	5	28.7%	30.2
P6	“The head of government of [X] is [Y].”	60	26.7%	88.6
P127	“[X] is owned by [Y].”	25	26.0%	153.2
P36	“The capital of [X] is [Y].”	60	20.0%	65.0
P137	“[X] is operated by [Y].”	25	19.3%	106.8

Table 5: Analytical per-fact-footprint upper bounds for the *known* facts at each scale, vs. our bit-budget estimate. KNOWLEDGE-NEURONS assumes $8 \cdot d_{\text{model}}$ params/fact; ROME assumes $d_{\text{model}} + d_{\text{inter}}$. Source: `results/per_fact_footprint.json`.

Model	Known facts	Params/fact (KN)	KN share	ROME share	Bit-budget share
PYTHIA-160M	812	6,144	3.07%	1.92%	$\sim 0.00022\%$
PYTHIA-410M	1088	8,192	2.20%	1.37%	$\sim 0.00022\%$
PYTHIA-1B	1145	16,384	1.85%	1.16%	$\sim 0.00010\%$
PYTHIA-2.8B	1443	20,480	1.06%	0.67%	$\sim 0.00006\%$

FFN value vector cannot “own” a single fact, otherwise the aggregate would exceed the bit-budget figure by the four-orders-of-magnitude shown here. This is consistent with the polysemantic-neuron picture of Geva et al. [2021] and the rising parameter specialization of Hong et al. [2025].

4.6 Surprises

Three observations were not predicted by the literature review.

TAXI accuracy plateaus near 1B. PYTHIA-410M and PYTHIA-1B have nearly identical TAXI accuracy (28.08% vs. 27.60%) despite a $2.5\times$ parameter increase. PYTHIA-2.8B only improves to 30.24%. Categorical knowledge appears to saturate at modest model sizes — consistent with Hong et al. [2025]’s finding that specialization (rather than additional capacity) drives knowledge utilization in larger models.

BEAR accuracy keeps growing. $6.1\% \rightarrow 8.9\% \rightarrow 9.7\% \rightarrow 13.0\%$ across the four sizes. Relational knowledge, which is more entity-specific, benefits more from scale than categorical knowledge.

Best-known relations are the most “standard.” P105 (taxonomic level), P6 (head of government), P36 (capital), P127 (ownership) are exactly the textbook examples — direct confirmation that the benchmark is dominated by the kind of standard knowledge our hypothesis targets, and that the parameter-share figures are not artefacts of obscure-fact querying.

5 Discussion

5.1 Why is the share so small?

Three mechanisms compress the apparent footprint to one part in a million.

Compositional storage. “Peanut $\rightarrow$ legume” generalizes to dozens of inherited properties (has-shell, contains-protein, edible, ...) without spending per-property bits. The bit complexity of the categorical edge is far smaller than the bit complexity of enumerating its entailments. TAXI probes exactly this structure — and the result shows up cleanly: TAXI achieves the highest per-fact accuracy (23–30%) but the lowest per-fact bit yield (small answer sets, low $\log_2 |\mathcal{A}|$).

Parameter sharing across facts. Geva et al. [2021] establish that a single FFN value vector activates for many concepts; Hong et al. [2025] quantify how this specialization rises with model generation. Our four-order-of-magnitude gap between the per-fact-footprint upper bound (ROME at 0.67%–1.92%) and the bit-budget estimate ($\sim 10^{-4}\%$) is the cleanest aggregate evidence we are

aware of that polysemantic value vectors are the dominant storage regime, not exclusive-ownership “knowledge neurons”.

Junk-data dilution. Allen-Zhu and Li [2024] show that 1:7 useful-to-junk training-data ratios cause $20\times$ capacity loss. PYTHIA is trained on THE PILE, which is dense in non-knowledge content (code, comments, dialogue), so most parameters serve linguistic competence and reasoning circuits rather than standard knowledge. The $\sim 5.56 \times 10^9$ bit budget at PYTHIA-2.8B is therefore an over-estimate of the budget actually allocated to knowledge — but it is the right denominator if we want to know “what fraction of the trained model is standard knowledge”.

5.2 What occupies the rest of the bit budget?

The 2.78B-parameter PYTHIA model has 5.56×10^9 bits of capacity at $\beta = 2$ bpp. Standard categorical and relational knowledge accounts for only 3.36×10^3 bits — a factor of 1.6×10^6 smaller. The plausible occupants of the remaining capacity, suggested by adjacent literature but not measured here, are:

- **Tail-of-distribution facts** about uncommon entities (people, places, events) that BEAR and TAXI exclude by construction. Mallen et al. [2023] show long-tail facts have very different accessibility profiles.
- **Linguistic competence and stylistic memorization** — a 2.8B PYTHIA on THE PILE reproduces enormous amounts of code, technical text, and forum dialogue.
- **Implicit reasoning circuits:** induction heads, copying mechanisms, arithmetic primitives, multi-hop chains.
- **Raw memorized passages** — Morris et al. [2025]’s 3.6 bpp ceiling is achieved on uniform-random strings, suggesting a large fraction of capacity is raw memorization rather than abstracted knowledge.

5.3 Implications

Distillation and quantization. If standard knowledge occupies $\sim 10^{-4}\%$ of the bit budget, the everyday-fact core is essentially free of capacity pressure. int4 quantization halves the β ceiling [Allen-Zhu and Li, 2024] but still leaves $\sim 10^4\times$ headroom for standard knowledge. The user-visible degradation reported in some quantization work [Hong et al., 2025] is therefore likely to come from long-tail facts, reasoning circuits, or linguistic competence, not from the standard-knowledge core.

Interpretability. The four-order-of-magnitude gap between per-fact-footprint upper bounds and the aggregate bit-budget figure quantifies a long-standing intuition: parameter sharing across facts must be the dominant regime in modern LLMs. Per-fact ROME edits work [Meng et al., 2022], but the *aggregate* of all such edits must massively over-count the bits actually used.

Training-data design. The slope from 160M to 2.8B in BEAR accuracy (6.1% to 13.1%) is gentle: standard-knowledge accumulation is not the binding constraint on model scale. Reasoning, stylistic, and long-tail accumulation must drive most of the capacity demand — a frame that points training-data design at long-tail entity coverage and reasoning chains, not at brute-force enumeration of textbook facts.

5.4 Hypothesis status

H1 (volume): supported. Standard knowledge consumes well under 1% of the bit budget, by four to five orders of magnitude. Even extrapolating to a benchmark-of-benchmarks $100\times$ larger than BEAR and TAXI combined would still leave the figure under 0.01%.

H2 (scaling): supported with caveat. The fraction *decreases* with model size: 410M $\rightarrow$ 1B halves it, 1B $\rightarrow$ 2.8B halves it again. Bootstrap CIs are disjoint, so the trend is statistically reliable. The caveat is the slope is gentle in absolute terms (-1.5×10^{-4} pp/decade); the dilution effect is real but modest.

H3 (typology): mixed. TAXI categorical accuracy (23–30%) consistently exceeds BEAR relational accuracy (6–13%) — the model knows categories better than entity-specific relations. But TAXI

contributes far fewer aggregate bits because it has $5\times$ fewer queries and smaller answer sets. Perfect density runs in the opposite direction (BEAR ~ 0.40 bits/fact vs. TAXI ~ 0.15 bits/fact). The hypothesis as originally stated — categorical knowledge is denser per fact — is wrong: relational facts carry more bits per query because the answer space is larger, even though the model knows fewer of them. Categorical knowledge is better-known but lower-bit; relational knowledge is harder but higher-bit.

5.5 Limitations

- **The probe is a lower bound on stored knowledge.** A model may contain knowledge it fails to express under a fixed template (paraphrase or surface-form sensitivity). Wiland et al. [2024] show LL-ranking is much more robust than cloze probing, but residual under-counting is still possible because we use only the first BEAR template per relation. Multi-template averaging would tighten the numerator.
- **The 2 bpp ceiling may not transfer to undertrained models.** PYTHIA is trained on 300B tokens — below the saturation regime where the 2 bpp law is tightest. Reporting also at $\beta = 3.6$ bpp partially addresses this; that denominator reduces the share by $\sim 45\%$, not enough to change conclusions.
- **Single architecture family.** All four models are PYTHIA (GPT-NeoX architecture). Generalization to LLaMA, Mistral, and Phi is plausible but not directly tested. Hong et al. [2025] specifically warn that specialization differs across families.
- **Coverage of “standard knowledge.”** BEAR and TAXI only partially cover commonsense, procedural (“how to bake bread”), causal (“X causes Y”), and temporal knowledge. A more comprehensive enumeration could push the share up by one or two orders of magnitude — but would still leave it well under 1%.
- **No quantization sweep.** Allen-Zhu and Li [2024] show int4 halves capacity. We do not measure how standard knowledge survives quantization vs. tail knowledge.
- **Bootstrap caveat.** The bootstrap 95% CIs are slightly narrower than they should be because we resample within (relation, fact) pairs but not across templates. Adding template-level bootstrap would widen the intervals modestly.

6 Conclusion

We asked what fraction of an LLM’s parameters is devoted to standard categorical and conditional knowledge — the kind a competent reader can state about peanuts, Georgia, or symphony orchestras. By combining log-likelihood probing on BEAR and TAXI with the 2 bits-per-parameter ceiling of Allen-Zhu and Li [2024], we measure the share at $6 \times 10^{-5}\%$ to $2 \times 10^{-4}\%$ across PYTHIA models from 160M to 2.8B parameters — roughly one part in a million of stored capacity. The fraction shrinks monotonically with model size from 410M onward; bootstrap 95% confidence intervals at 1B and 2.8B are disjoint from those at smaller scales, so the decay is statistically reliable. The mechanism is sub-linear scaling of acquired bits (BEAR slope 0.54, $p = 0.04$) against linear growth of capacity.

Key takeaway. Standard knowledge is not the binding capacity constraint at modern scales: by four to five orders of magnitude, the bit budget is spent on something else — long-tail entities, linguistic competence, reasoning circuits, raw memorization. Aggressive quantization or distillation is essentially free for the standard-knowledge core; the user-visible degradation that practitioners report under such transformations must come from elsewhere.

Future work. Five concrete follow-ups extend the measurement.

- **Architecture transfer.** Replicate on Llama-3.2-{1B, 3B} and Mistral-7B to test whether the one-part-in-a-million figure is PYTHIA-specific. Hong et al. [2025] motivate this directly.
- **Multi-template averaging.** Re-run with 5–8 templates per BEAR relation and confidence intervals over template variance, to tighten the numerator and address the lower-bound caveat.
- **Quantization sweep.** Run the analysis at fp16, int8, and int4 to test whether standard facts decay faster or slower than long-tail facts under capacity halving.

- **Per-fact causal localization.** Combine ROME-style causal localization with our known-fact list to compare the *measured* per-fact footprint with the analytical KNOWLEDGE-NEURONS and ROME bounds, closing the four-order-of-magnitude gap empirically.
- **Broader knowledge benchmarks.** Add commonsense (CommonsenseQA, ATOMIC), procedural, and temporal benchmarks to test whether the share remains well under 1% once the definition of standard knowledge is stretched to its broadest reading.

We release probing code, per-relation breakdowns, and bootstrap CIs so that all numbers in this paper reproduce from the project repository. The goal is not to settle the question of where LLM capacity goes — that requires the long list of follow-ups above — but to fix one number that prior work had not fixed: standard knowledge is roughly one part in a million.

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